

[79]

SARDAR PATEL UNIVERSITY
BCA SEMESTER – II

DATE : 14-8-2019

TIME : 2:00 P.M to 5:00 PM

Total Marks : 70

[10]

- [20]

1. Define: indirection operator, pointer variable.
2. Differentiate between '*' and '&' operators in pointers.
3. List out benefits of pointer.
4. Differentiate: structure and union.
5. What is structure definition? Write the importance of tag in structure definition.

(P.T.O.)

6. List file modes available to manage the file in C.
7. Give the example of Primitive Data Structure.
8. What do you mean Linear Data Structure?
9. State various Applications of Stack.
10. What is a doubly Linked list?
11. State various types of queue.
12. Give representation of a Queue data structure.

- Q.3 [A] Define pointer variable. How can we declare and initialize pointer variable? How can we access value of variable through pointer type variable? [06]
 [B] Write note on: pointer to pointer. [04]
 OR
- Q.3 [A] Explain pointer arithmetic with example. [06]
 [B] Write a note on Dynamic memory allocation. [04]
- Q.4 [A] Explain fprintf and fscanf functions with example. [06]
 [B] Explain getw and putw functions with example. [04]
 OR
- Q.4 [A] What is structure? Explain its definition, declaration and assigning values to members of structure. [06]
 [B] Explain array within structure using suitable example. [04]
- Q.5 [A] Explain linear and nonlinear data structures. [06]
 [B] Describe classification of data structure. [04]
 OR
- Q.5 [A] Define Stack. Write an algorithm for peep and change operations of a stack. [06]
 [B] Write down the advantages of data structure. [04]
- Q.6 What is queue? Write algorithms to insert an element into a simple queue and delete an element from a simple queue. [10]
 OR
- Q.6 Define linked list. Write algorithms to insert new element at the beginning of a singly linked list and to insert new element at the end of a singly linked list. [10]

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